Aditya Saini

908-912-6974 | adityasaini0328@gmail.com | linkedin.com/in/adityavsaini | github.com/Saini-01 | sunnyv.surge.sh

Education

Purdue University West Lafayette, IN

B.S. Computer Science (Systems Programming) *Expected Graduation: May 2029*

Relevant Coursework: Computer Architecture, Programming in C, Data Structures & Algorithms, Object-Oriented Programming, Discrete Mathematics, Linear Algebra, Multivariate Calculus

Experience

Embedded Engineer Sep. 2025 – Present

Purdue Electric Racing *West Lafayette, IN*

- Designed and implemented the driver-facing dashboard UI for the Electric Racing competition vehicle on a Nextion HMI display using Nextion Editor, authoring all screen graphics for live telemetry, fault codes, and sensor data
- Wrote embedded C on an STM32F4 (ARM Cortex-M4) driving 6+ dashboard screens and button-mapped page transitions over UART, mapping CAN fault codes to prioritized driver alerts
- Rewrote the display command queue from fixed-width slots to a packed byte-level ring buffer, eliminating padding on the UART stream and fixing intermittent dropped commands on the Nextion dashboard display

Digital Design Engineer July 2026 – Present

SoCET VIP, Purdue Electrical & Computer Engineering *West Lafayette, IN*

- Designed a byte-addressed memory copy controller (simple DMA) in SystemVerilog as a five-state Moore FSM that reads from a source address and writes to a destination address when a start signal goes high
- Split combinational next-state and output logic from a clocked state register with asynchronous reset, buffering read data in a register so it held stable across the read-to-write cycle
- Verified copy correctness, memory handshake timing, and reset behavior with a SystemVerilog testbench
- Developing a Space Invaders game on an FPGA in SystemVerilog (in progress), building modular synchronous RTL for real-time graphics rendering and game mechanics with simulation and FPGA validation
- Designed a VGA timing controller for 640×480 at 60 Hz from a 25 MHz pixel clock, using parameterized 10-bit horizontal and vertical counters to generate active-low HSYNC/VSYNC pulses and detect the visible area
- Timed each frame across 800 clock cycles per line and 525 lines including blanking intervals, generating pixel coordinates and a frame-tick signal on vertical blanking to sync game-state updates with display refresh

Build President & Programming Captain September 2021 – August 2025

FRC Team 2554 - The Warhawks *Edison, NJ*

- Led 60+ students as Build President to District Championships after 18 years, earning 3 regional awards.
- Built a CUDA-accelerated OpenCV vision pipeline in Python for AprilTag detection, improving autonomous positioning accuracy by 66%, leading to a larger scoring volume with greater scoring efficiency.
- Diagnosed and resolved cross-system hardware failures across CAN bus-chained motor controllers and sensors
- Designed robot subsystems in Onshape through successive prototyping cycles, including the intake and drivetrain

Projects

Custom 30-Button Gamepad | *Embedded C, PSoC, USB* January – April 2025

- Fabricated a custom 30-button gamepad on a PSoC microcontroller, writing embedded C firmware and a USB polling loop so the FRC Driver Station read it as a standard game controller and relayed inputs to the robot
- Improved driver control efficiency by 50% by giving every robot function its own dedicated button

Natural Language Shell | *C, POSIX, Python, Bash* June – August 2025

- Built NLSH, a custom Unix shell in C with process management, input tokenization, and shell built-ins (cd, help, exit), handling the full read-parse-execute loop rather than wrapping an existing shell
- Bridged the C core to a Python helper over popen for command translation, added OS-aware prompt context, and shipped a Bash installer that puts the shell on PATH

Technical Skills

Languages: C, C++, SystemVerilog, Verilog, RISC-V assembly, Python, Java, SQL, Bash
Systems: Linux, POSIX, process management, embedded C, RTOS, Docker, Makefiles, GCC, GDB
Embedded/Hardware: STM32 (Cortex-M4), STM32CubeIDE, PSoC, Jetson Orin Nano, UART, CAN, SPI, I2C
EDA/Tools: Cadence Genus/Innovus, AWS, OpenLane, LTspice, FPGA synthesis/timing, RTL verification, CUDA, Git